

Human-Centered AI Journal Research and Design

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Executive Summary

Artificial Intelligence is becoming increasingly integrated into applications that support everyday activities such as learning, productivity, healthcare, and personal wellbeing. While these technologies offer significant opportunities to improve user experiences, they also introduce new challenges regarding trust, transparency, privacy, and user autonomy. These challenges become even more important when AI is incorporated into emotionally sensitive activities such as journaling, where users often record personal thoughts, emotions, and life experiences.

This project explores how Human-Centered Artificial Intelligence (HCAI) can be applied to the design of an AI-assisted journaling platform. Rather than investigating how AI can automate journaling, the objective was to understand how AI can meaningfully support users without reducing their control over personal reflection.

To achieve this, a research-driven approach was adopted. Existing journaling applications were first reviewed to identify common limitations, followed by a structured user study designed to understand current journaling habits, expectations from AI, privacy concerns, and preferred interaction models. The collected feedback became the foundation for designing **EchoJournal**, a prototype that demonstrates how AI can assist users while remaining transparent, ethical, and supportive.

Instead of positioning AI as an emotional advisor, EchoJournal treats AI as a reflection assistant that helps users organize thoughts, identify long-term patterns, and encourage consistent journaling while leaving interpretation and decision-making entirely to the user.

The outcome of this project is not simply a redesigned journaling interface, but a research-backed demonstration of how Human-Centered AI principles can guide the development of trustworthy AI-assisted wellbeing applications.

Project Background

Digital journaling has evolved from handwritten notebooks to mobile applications that allow users to record experiences, monitor habits, and reflect on personal growth. More recently, advances in generative AI have encouraged developers to introduce features such as writing assistance, emotional summaries, personalized prompts, and mood tracking.

Although these capabilities appear promising, they also raise important questions. Users may appreciate AI-generated suggestions, but they may not want AI to interpret their emotions, make assumptions about their mental state, or influence personal decision-making. The challenge is therefore not whether AI should be used in journaling, but how it should interact with users in a way that feels trustworthy, respectful, and genuinely helpful.

Many existing AI-powered wellbeing applications focus heavily on automation while giving comparatively little attention to user control or transparency. Features such as automated emotional analysis or AI-generated advice may be technically impressive, yet they can unintentionally reduce user confidence if individuals do not understand how conclusions are produced or how their personal information is being used.

This project approaches the problem from a different perspective. Rather than beginning with AI capabilities, it begins with people. Understanding how users currently journal, what difficulties they experience, and how they expect AI to behave became the starting point for every design decision made throughout the project.

Project Objectives

The primary goal of this project was to investigate how Human-Centered AI principles can be incorporated into the design of an AI-assisted journaling platform that prioritizes user trust, autonomy, and well-being.

To achieve this objective, the project focused on four key activities:

- Understanding current journaling behaviours and challenges through user research.
- Investigating user expectations regarding AI-assisted journaling, including privacy, transparency, and personalization.
- Translating research findings into practical design requirements.
- Developing a high-fidelity prototype that demonstrates responsible AI interaction within a digital journaling environment.

Unlike traditional design projects that prioritize interface improvements alone, this work emphasizes how research findings can directly influence the behaviour of AI within an application.

Research Methodology

The project followed a research-first methodology in which user insights were collected before any design work began. This ensured that the final prototype addressed genuine user needs rather than assumptions made during development.

The overall workflow consisted of five stages.

Existing Journal App Review



User Research Planning



Google Form Survey



Response Analysis



EchoJournal Prototype Design

The first stage involved reviewing existing digital journaling applications to understand their strengths and limitations. While many platforms successfully supported basic journaling activities, several recurring issues became apparent. Applications often provided limited personalization, inconsistent long-term engagement, and AI features that felt either overly generic or insufficiently transparent. These observations helped define the research questions that would later be explored through the user survey.

A structured Google Form was then designed to collect both quantitative and qualitative feedback from participants. Rather than concentrating only on interface preferences, the questionnaire explored broader topics including journaling habits, trust in AI, privacy expectations, emotional comfort, desired AI features, and concerns surrounding AI-generated insights.

The responses were reviewed using thematic analysis to identify recurring patterns across participants. Instead of evaluating each survey question independently, related responses were grouped into broader themes such as trust, personalization, privacy, emotional safety, and user control. These themes became the primary design requirements that guided the development of EchoJournal.

By placing research before design, every major feature included in the prototype could be directly linked to a user need identified during the study. This approach ensured that the resulting application remained grounded in Human-Centered AI principles rather than technological novelty.

User Research and Key Findings

The user study served as the foundation of this project by providing direct insight into how individuals currently engage with digital journaling and what role they expect Artificial Intelligence to play within that experience. Rather than assuming which features users would find valuable, the survey was designed to understand their existing habits, expectations, concerns, and willingness to interact with AI in emotionally sensitive situations.

A structured Google Form containing **29 questions** was developed to collect both quantitative and qualitative feedback. The questionnaire was organized into multiple sections, beginning with participants' journaling habits before gradually introducing questions related to AI-assisted journaling, personalization, privacy, emotional wellbeing, and trust. This progression allowed participants to first reflect on their existing experiences before evaluating how AI could improve or potentially interfere with those experiences.

Instead of focusing only on interface preferences, the questionnaire explored broader research questions such as:

- Why do people maintain journals?
- What difficulties do users experience with existing journaling applications?
- What type of AI assistance feels genuinely useful?
- How comfortable are users sharing personal reflections with AI?
- What level of control should users have over AI-generated insights?
- Which features would encourage long-term journaling engagement?

The survey combined multiple-choice questions, Likert-scale ratings, and open-ended responses, allowing participants to provide both measurable feedback and detailed personal opinions. This combination proved particularly valuable because it not only identified common trends but also explained the reasoning behind those trends.

The responses revealed a clear pattern: participants were generally positive about integrating AI into journaling, but only when AI functioned as an assistant rather than an authority. Most participants appreciated the idea of receiving writing prompts, summaries,

reminders, and long-term insights, yet they were uncomfortable with AI making assumptions about emotions or providing psychological advice. This distinction became one of the most influential findings of the entire study.

Privacy emerged as another recurring concern throughout the survey. Participants expressed hesitation about storing highly personal journal entries within AI-powered applications without understanding how their information would be processed or used. Many indicated that they would be more willing to adopt such applications if AI interactions were transparent and remained under user control. Rather than viewing privacy as a technical feature, participants associated it directly with trust in the overall system.

The survey also highlighted the importance of personalization. Users did not want identical experiences or generic AI-generated responses. Instead, they expected the application to adapt to their individual journaling habits through personalized reminders, customizable themes, mood tracking, and long-term reflection. However, participants emphasized that personalization should always remain configurable rather than automatic, reinforcing the need for user autonomy throughout the application.

Another interesting observation involved accessibility and interaction preferences. While traditional text-based journaling remained the preferred method for most participants, many believed that voice journaling would make reflection easier during situations where typing felt inconvenient or emotionally difficult. This suggested that supporting multiple interaction methods could improve both accessibility and long-term engagement without fundamentally changing the journaling experience.

Rather than treating these findings as isolated observations, they were consolidated into a set of functional design requirements. Every major feature included within EchoJournal can be directly linked to one or more research findings collected during the survey.

Research Finding	Design Decision in EchoJournal
Users wanted AI assistance without losing control	AI-generated prompts and recommendations remain optional throughout the application.
Privacy strongly influenced trust	Privacy settings and user-controlled AI interactions were incorporated into the design.
Participants preferred personalized experiences	Personalized reminders, customizable themes, and adaptive recommendations were introduced.

Research Finding	Design Decision in EchoJournal
Long-term reflection was considered valuable	Mood trends, journaling history, and insight dashboards were added.
Users requested multiple interaction methods	Both text-based and voice journaling were supported within the prototype.
Transparency increased confidence in AI	AI-generated content is clearly distinguishable from user-created journal entries.

One of the most valuable outcomes of the research was the realization that users were not asking for more intelligent AI; they were asking for more trustworthy AI. Participants consistently favored systems that respected personal boundaries, explained their role clearly, and enhanced reflection without attempting to replace it. This insight became the central design philosophy of EchoJournal and shaped every major decision made during the development of the prototype.

From Research to Design

The findings collected during the user study became the foundation for the design of **EchoJournal**. Rather than adding AI features based on current technology trends, every major component of the prototype was designed to address a specific user need identified during the research process. This ensured that the application reflected Human-Centered AI principles in practice rather than treating them as abstract design guidelines.

The completed prototype represents a digital journaling platform where Artificial Intelligence functions as a supportive companion throughout the reflection process. Instead of replacing personal writing or attempting to interpret emotions, AI is integrated in ways that reduce friction, encourage consistency, and help users better understand their own journaling habits over time.

The prototype consists of several interconnected modules that together support the complete journaling experience, including journal creation, voice journaling, historical entries, personalized insights, reminders, goal tracking, calendar integration, and application settings. While each module serves a different purpose, they all share the same design philosophy: AI assists the user, while the user remains in complete control of the journaling experience.

One of the most significant design decisions involved the role of AI during journal creation. Survey participants appreciated the idea of receiving writing prompts and suggestions but did not want AI to write journal entries on their behalf. To address this concern,

EchoJournal provides optional AI-generated prompts that help users begin or continue writing without influencing the content of their reflections. Users can ignore these suggestions entirely, request additional prompts, or write independently, ensuring that the journal always remains a personal record of their own thoughts.

Another recurring theme throughout the research was accessibility. Although traditional text-based journaling remained the preferred method for many participants, several respondents indicated that they would appreciate alternative methods of recording their thoughts. As a result, voice journaling was incorporated into the prototype as an additional input option. Rather than replacing written entries, voice recording provides greater flexibility for users who may find speaking more natural during emotionally significant moments or when typing is inconvenient.

Long-term reflection also emerged as an important requirement during the survey. Participants expressed interest in recognizing patterns within their journal entries rather than simply storing daily records. To support this objective, the prototype includes an insights dashboard that summarizes journaling activity through mood trends, writing consistency, and historical reflections. These visualizations are designed to encourage self-awareness by presenting information in an organized and accessible format. Importantly, the dashboard avoids drawing conclusions about the user's emotional wellbeing. Instead, it provides observations that users can interpret for themselves, reinforcing the principle that AI should support reflection rather than replace human judgement.

Personalization became another key design consideration. Survey responses indicated that users wanted the flexibility to adapt the application to their own preferences instead of receiving a fixed experience. EchoJournal, therefore, includes configurable reminders, customizable themes, journaling goals, and personalized recommendations that evolve according to user preferences. Unlike many AI systems that automatically modify user experiences, these personalization features remain entirely configurable, allowing users to decide how much AI involvement they prefer.

Throughout the design process, particular attention was given to maintaining clear boundaries between AI-generated content and user-generated content. AI recommendations are intentionally presented as suggestions rather than instructions, and users retain the ability to modify, ignore, or disable AI assistance at any point. This distinction helps preserve user autonomy while reducing the risk of over-reliance on automated recommendations.

The final prototype demonstrates that Human-Centered AI extends beyond interface design. It influences how AI communicates, how much control users retain, and how trust is established through transparency and responsible system behaviour. Rather than designing technology that attempts to think on behalf of users, EchoJournal was designed to encourage meaningful self-reflection while allowing AI to quietly support the process in the background.

The complete prototype is illustrated in Figure 1, showcasing the major application screens developed from the research findings, including the dashboard, journal interface, voice journaling, insights, calendar, reminders, goals, and settings. Together, these components represent a unified AI-assisted journaling experience grounded in user research and Human-Centered AI principles.

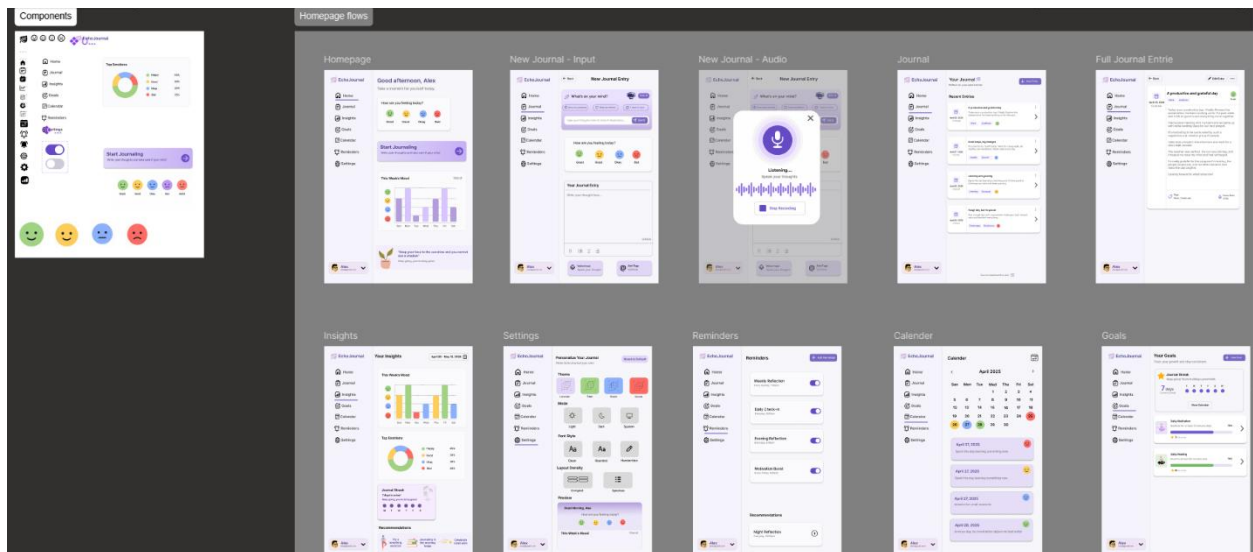


Figure 1. High-fidelity EchoJournal prototype developed in Figma, illustrating the primary user workflow and AI-assisted journaling features.

Discussion

One of the most significant outcomes of this project was recognizing that users are not necessarily looking for more advanced AI; they are looking for AI they can understand and trust. Throughout the survey, participants consistently supported the idea of AI-assisted journaling, but only when the technology respected personal boundaries and remained under the user's control. This observation shifted the focus of the project from developing intelligent features to designing responsible interactions.

The research also demonstrated that user expectations extend beyond functionality. Participants valued transparency as much as personalization, and privacy often influenced their willingness to adopt AI-powered journaling applications. These findings reinforced the importance of treating Human-Centered AI as a practical design framework rather than simply an ethical guideline. By incorporating user feedback into every stage of development, the final prototype reflects not only technical capabilities but also the values and expectations of its intended users.

Another important takeaway was the relationship between AI assistance and human autonomy. Features such as writing prompts, journal summaries, mood trends, and personalized reminders were positively received because they supported the journaling process without influencing personal interpretation. This balance between assistance and independence became one of the defining characteristics of EchoJournal and demonstrates how AI can enhance human experiences without replacing them.

Although the prototype successfully demonstrates the application of Human-Centered AI principles, several limitations should be acknowledged. The project focused on designing and evaluating a high-fidelity prototype rather than developing a production-ready application. Consequently, the AI interactions presented within the prototype represent intended system behaviour rather than fully implemented functionality. Additionally, the research findings were derived from a limited participant group and therefore may not fully represent the expectations of a broader user population. Expanding the participant pool and evaluating the prototype through longitudinal user studies would provide valuable insights into long-term engagement, trust, and usability.

Future work could extend this project by implementing a functional version of EchoJournal and integrating modern language models to generate context-aware journaling assistance. However, any future implementation should continue to prioritize transparency, explainability, user consent, and privacy. Features such as explainable AI, on-device processing, encrypted journal storage, multilingual support, and accessibility enhancements could further strengthen the application while remaining consistent with Human-Centered AI principles.

Conclusion

This project explored how Human-Centered Artificial Intelligence can be applied to the design of an AI-assisted journaling platform that prioritizes trust, transparency, privacy, and user autonomy. Rather than beginning with technology, the project began by understanding people—their journaling habits, expectations, concerns, and perceptions of AI. A structured user study provided the evidence required to transform these insights into

practical design requirements, ultimately leading to the development of the EchoJournal prototype.

The completed prototype demonstrates that AI can meaningfully support personal reflection without becoming intrusive or replacing human judgement. Features such as optional writing prompts, voice journaling, personalized reminders, mood insights, and progress tracking were intentionally designed to complement the user's journaling experience while ensuring that control always remains with the individual. Throughout the project, Human-Centered AI principles guided not only interface design but also the behaviour and responsibilities assigned to AI within the system.

More broadly, this project highlights that the success of AI-powered applications depends not only on the intelligence of their algorithms but also on the quality of the interaction they create with users. Designing AI systems that are transparent, respectful, and aligned with human values is essential for building long-term trust, particularly in applications that involve sensitive personal information. EchoJournal represents one possible approach to achieving this balance by demonstrating how user research and ethical design principles can be integrated into the development of responsible AI-assisted experiences.

The project has strengthened practical skills in Human-Centered AI, user research, survey design, qualitative analysis, requirements engineering, ethical AI, and interaction design. More importantly, it demonstrates the value of allowing research—not assumptions—to drive the development of AI systems that genuinely support human needs.